IoT Access Control

Mid-Quarter Progress Report

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1 ABSTRACT

2 INTRODUCTION

3 RELATED WORK

4 PROJECT DESCRIPTION

5 TIMELINE

5.1 Progress to Date

6 RESULTS AND ANALYSIS

6.1 Experimental Setup

We are working in a compartmentalized environment, which allows us to develop the server (MQTT broker) and clients (simulated publishing and subscribing devices) in relative isolation. The clients are in Docker containers, allowing them to be instantiated identically as many times as needed in the experimental environment, each running inside its isolated container but able to send and receive network traffic with the server. This setup allows each team member to separately develop the server and client components away from the actual test environment as well, allowing development and debugging in a way that does not interfere with running experiments.

The server used to perform all the experimental runs, and upon which our performance benchmarks is based, is an Intel® Core™ i9-10980XE (36 cores) running at 4.80 GHz configured with 128 Gb of memory. It was running the Arch-based Garuda Linux distribution 6.13.7 Zen Linux kernel at the time of the experimental results reported here.

7 CONCLUSIONS

8 INDIVIDUAL CONTRIBUTIONS

9 REFERENCES